RFID-based Access Control System with Attendance Tracking

*A **MINI / MINOR / MAJOR** Project Report submitted in partial fulfillment of the requirements for the award of the degree of*

BACHELOR OF TECHNOLOGY

in

ROBOTICS AND AUTOMATION

BY

ROHIT PAUL

UID: TNU2020036100006

Under the supervision of

Anirban Dutta

Think Again Lab, West Bengal, India



DEPARTMENT OF ROBOTICS & AUTOMATION ENGINEERING

THE NEOTIA UNIVERSITY

D.H. ROAD, SOUTH 24-PARAGANAS, WEST BENGAL, INDIA, PIN-743368

MAY 2024

Certificate

We hereby recommend that the Project entitled ***"RFID-based Access Control System with Attendance Tracking"*** worked under our guidance may please be accepted in the partial fulfillment of the requirement for the degree of "Bachelor in Technology" in Robotics and Automation Engineering of 'The Neotia University.' The project report in our opinion is worthy for its acceptance. During the work '**ROHIT PAUL**' was found to be sincere, regular, and hardworking and has successfully completed the thesis work assigned to him.

Anirban Dutta
Name of the project guide
(Project Guide)

…….

(HOD, R&A)
The Neotia University

Declaration of Originality and Compliance of Academic Ethics

I hereby declare that this report contains literature survey and original research work done by the under signed candidate/s, as part of my **"Bachelor in Technology Studies"**.

All information in this document have been obtained and presented in accordance with academic rules and ethical conduct.

I also declare that, as required by these rules and conduct, I have cited and referenced all materials that are not original to this work.

Project Title: RFID-based Access Control System with Attendance Tracking

Student

Name: ROHIT PAUL

UID: TNU2020036100006

Signature with Date:

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to all those who provided me with the possibility to complete this final year project. First and foremost, I am profoundly thankful to my project supervisor, Anirban Dutta, whose expertise, understanding, and patience added considerably to my graduate experience. Your invaluable guidance and continuous encouragement have been instrumental in the completion of this project.

I am also grateful to the faculty and staff of the Robotics and Automation at The Neotia University for their unwavering support and for providing an excellent environment for learning and research.

I wish to extend my sincere thanks to my fellow classmates and friends, whose camaraderie and support made the challenging moments more bearable. Your insightful discussions and collaborative spirit greatly enhanced the quality of this work.

Furthermore, I would like to acknowledge the assistance of Think Again Lab, which provided me with the necessary resources and data, making this project possible. The practical insights gained from this collaboration have been invaluable.

I am deeply indebted to my family for their love, support, and understanding throughout my academic journey. Your belief in my abilities has been my greatest strength.

Finally, I would like to thank anyone else who contributed, directly or indirectly, to this project. Your assistance and encouragement have been a source of inspiration.

Thank you all.

ROHIT PAUL

Place: The Neotia University

Date:

Contents

1. Chapter 1 – Introduction
2. Chapter 2 – Literature review
3. Chapter 3 – Methodology / Phases of Software
4. Chapter 4 – Hardware and Software Setup
5. Chapter 5 – Results and Discussions
6. Chapter 6 – Conclusion and Future Scope
7. References

LIST OF FIGURES

Chapter 4:

ESP32 DEVKITV1 Pinout
Block Diagram of MFRC522
Detailed Block Diagram of MFRC522
Pinning configuration HVQFN32 (SOT617-1)
MFRC522 Read/Write mode
RFID MFRC522 MODULE
Keypad 4x4 Membrane Module
Pinning Diagram of Keypad
Solenoid Lock
Dimensions of Solenoid Lock
LCD Module with Proper Dimensions
16x2 LCD Pinout
Relay Module
Working Principle of Relay Module
Circuit Diagram of System

Chapter 5:

Initial Setup
Card Scanning in RFID
Lock system working image
Final Setup

LIST OF TABLES

Chapter 4:

Features of ESP32 DEVKITV1
Pin Description of 16x2 LCD
Buzzer Specification
Pin Configuration of the system

Chapter 5:

Performance Test data
Average Card read distance
Performance test data for password

ABSTRACT

Security and attendance management are paramount concerns in various sectors, including education and business. This project addresses these challenges by developing a comprehensive RFID and password-based attendance and lock system with cloud integration, leveraging the ESP32 DEVKITV1 development board as the core platform.

The proposed system, termed RFID and Cloud-Integrated IoT Attendance and Lock System, aims to streamline attendance tracking and access control processes. Through RFID card authentication and password input, users can register attendance and unlock doors securely. Integration with cloud services, specifically Google Sheets, facilitates real-time storage and management of attendance records.

Key objectives of the project include:

- Designing and implementing an RFID and password-based attendance registration and door access control system.
- Integrating cloud connectivity using Google Sheets for storing and managing attendance records.
- Ensuring compatibility and usability of the system across different environments and user scenarios.
- By combining RFID technology, password authentication, and cloud integration, the project aims to contribute to improved security, efficiency, and convenience in attendance management and access control systems.

Keywords: RFID, ESP32 DEVKITV1, IoT, Cloud Integration, Attendance Management, Access Control

INTRODUCTION

In today's rapidly evolving technological landscape, security and attendance management have become critical aspects across various domains, including education, corporate, and industrial sectors. As organizations strive to enhance efficiency, accuracy, and security in their operations, the integration of innovative technologies has emerged as a promising solution. This project endeavors to address the challenges associated with traditional attendance management and access control systems by developing a sophisticated RFID and password-based attendance and lock system with cloud integration.

The project capitalizes on the capabilities of the ESP32 DEVKITV1 development board, a versatile platform known for its robustness and flexibility in IoT applications. By leveraging this platform, along with RFID technology and cloud services, the project aims to deliver a comprehensive solution that streamlines attendance tracking and door access control processes.

Through the implementation of RFID card authentication and password input mechanisms, users will be able to register attendance and unlock doors securely and conveniently. Furthermore, the integration of cloud services, particularly Google Sheets, will enable real-time storage and management of attendance records, enhancing accessibility and scalability.

This introduction sets the stage for the project's objectives and highlights the significance of leveraging cutting-edge technologies to address contemporary challenges in security and attendance management. As the project progresses, it endeavors to contribute to the advancement of efficient, secure, and user-friendly solutions that cater to the evolving needs of modern organizations.

Safeguarding homes has become a concerning topic, with increasing threats of burglary. Home security necessitates a secure door system. The aim of this work is to address security issues prevalent in today's world, where ensuring effective security solutions remains challenging even in technologically advanced environments. The rise of the Internet of Things (IoT) has introduced revolutionary effects in many cities, prompting the proposal of a Smart door lock system.

The Smart door lock system operates with a card and an RFID Reader box. By placing the card in front of the RFID Reader box, the lock unlocks with a beep. This RFID Door Lock can be easily constructed and installed on any door. It functions as an electrically operated door lock that opens when a voltage (typically 12v) is applied to it. This system ensures that only authorized individuals can access the doors, as it utilizes RFID technology to differentiate between valid and invalid users.

The system allows for remote monitoring from anywhere in the world, providing continuous updates on the door's status. It enables authorized individuals to access the doors, allowing for remote house monitoring. Additionally, a digital door locking system is implemented and governed by an RFID reader, which authenticates and validates users, automatically opening the door. It also maintains a record of user check-ins and check-outs.

Authentication of users before entering a secure space is crucial, and RFID technology provides this solution. The system offers fast, secure, and convenient check-in and check-out conditions. It includes a door locking system that opens when the user's tag is in contact with the reader, matching the user information stored in the database. RFID technology controls the opening and closing of the door, providing a secure access solution while maintaining user records.

Providing a robust security system for a room is crucial for protecting valuable items and preventing unauthorized access. Therefore, security measures must be enhanced to deter theft or data breaches. In this context, an electronic locking system is designed to improve security and authentication, with a focus on automating door lock security.

Traditionally, people have relied on conventional keys for security, but these can be vulnerable to criminal activity. Conventional door lock systems operate mechanically, requiring the insertion of the correct key and rotation to open the door. However, this method lacks reliability and is susceptible to theft.

To address these shortcomings, digital systems utilizing Radio Frequency Identification (RFID) technology have been introduced. RFID cards replace conventional keys and offer enhanced security by restricting access to authorized users. These cards are practical and thin, resembling automated teller machine cards. However, RFID technology has its limitations, including sensitivity to radio waves and potential data loss, rendering the card unusable.

Several research papers explore advancements in door lock systems. One focuses on a home system based on Bluetooth technology, while another explores the integration of RFID for security and automatic lighting control. Monitoring door status, controlling access, and enhancing security authentication in homes are also areas of investigation. Smart home systems incorporating digital door locks with interfaces such as keypads for inputting personal identification numbers are being developed. Additionally, there are proposals for enhancing the usability, scalability, and security of RFID-equipped Internet of Things (IoT) devices.

LITERATURE REVIEW

In SMART DOOR LOCK SYSTEM By Nayana R and Shashidhar R, the paper outlines a system where fingerprints of authorized individuals are stored in a microcontroller. A matching algorithm is used to verify authorization, and if successful, an OTP (One Time Password) is sent to the individual's mobile number via GSM. If an unauthorized attempt is made to access the door, a buzzer alerts of the intrusion. This system aims to provide high security at a low cost, applicable in sectors like banks and offices.

In SMART DOOR LOCK AND LIGHTING SYSTEM By Rahul Satoskar and Akarsh Mishra, an IoT-based smart door lock and lighting system for smart homes is proposed. It utilizes a digital password for door access and allows users to control and monitor home environment remotely. The system also incorporates smart lighting that can be controlled over the Internet.

ANDROID BASED SMART DOOR LOCKING SYSTEM By Manish Kumar, M Hanumanthappa, T V Suresh Kumar, and Amit Kumar Ojha presents a system where users can unlock the door with a pre-decided password. The system provides flexibility for password change/reset and temporarily disables after a certain number of incorrect attempts, enhancing security.

PASSWORD BASED SECURITY LOCK SYSTEM By Arpita Mishra, Siddharth Sharma, Sachin Dubey, and S. K. Dubey focuses on a password-based door security system. Users input a password via keypad, and upon correct entry, a motor opens the door lock. The system allows for adding new users and changing passwords via the keypad, with message display facilitated by an LCD module.

The integration of RFID technology with biometrics enhances security control systems, as demonstrated by Umar’ s work on an RFID-based security control system installed in hostels at the Punjab University campus. This innovative system not only captures the user's RFID card UID but also captures their image, which is then scanned and compared with a database for matching. If the card UID matches with the captured image, access is granted; otherwise, access is denied, triggering an alert alarm for security purposes. This combined approach significantly strengthens access control measures by incorporating both RFID authentication and biometric verification.

This system serves a pivotal role as both an entrance monitoring controller and an exit monitoring controller, making it ideal for deployment at entrance and exit gates. Its application in hostels underscores its effectiveness in enhancing security for residential environments. Moreover, to further improve system efficiency, response time can be optimized through the utilization of controller processes and real-time image processing techniques.

By merging RFID technology with biometrics, this advanced security control system offers heightened levels of authentication and access control. It represents a proactive approach to security management, leveraging the strengths of both RFID and biometric technologies to ensure robust protection against unauthorized access.

Radio Frequency Identification (RFID) technology has traversed a remarkable journey since its inception, shaping modern information systems and revolutionizing various industries. This comprehensive literature review delves into the historical development, current applications, and potential security and privacy implications of RFID technology, offering insights into its evolution and impact on the digital age.

The historical roots of RFID technology can be traced back to the early 20th century, with the development of radio technology and its application in military Identification Friend or Foe (IFF) systems during World War II. These early instances of RFID faced security vulnerabilities, with notable incidents such as Allied planes being compromised due to signal jamming and interception. However, these challenges spurred innovation, leading to the development of cryptographic solutions and challenge-response identification techniques to enhance security.

Over time, RFID technology has evolved from its military origins to find widespread application across various sectors. In contemporary times, RFID is integral to supply chain management, enabling real-time tracking of goods and optimizing inventory management processes. Additionally, RFID technology has found applications in access control systems, automatic payment systems, and asset tracking, driving efficiency and convenience in everyday transactions.

Despite its myriad benefits, RFID technology also raises significant security and privacy concerns. The ability to wirelessly identify and track objects without line-of-sight requirements poses risks of unauthorized access and data breaches. Concerns about corporate espionage, behavioral profiling, and mass surveillance have been raised by activists and privacy advocates, highlighting the need for robust security measures and regulatory oversight.

Looking ahead, the future of RFID technology holds promise for further innovation and advancement. As RFID systems become more pervasive and interconnected, addressing security vulnerabilities becomes paramount. Cryptographic techniques, intrusion detection systems, and regulatory frameworks will play a crucial role in mitigating risks associated with RFID technology. Moreover, interdisciplinary collaboration between academia, industry, and policymakers will be essential in shaping the future trajectory of RFID technology and ensuring its responsible use.

Radio Frequency Identification (RFID) technology has traversed a dynamic trajectory, evolving from its nascent stages to becoming a cornerstone of modern information systems. This extensive literature review delves deeper into the multifaceted aspects of RFID technology, exploring its historical evolution, contemporary applications, and the pervasive concerns surrounding its security and privacy implications.

RFID technology has its roots in the early 20th century, where it found its initial applications in military settings, particularly in Identification Friend or Foe (IFF) systems during World War II. However, these early incarnations of RFID were fraught with vulnerabilities, leading to instances of compromised security and interception. The exigencies of wartime necessitated rapid innovation, driving the development of cryptographic solutions and challenge-response identification techniques to fortify RFID systems against malicious exploitation.

The subsequent decades witnessed a gradual expansion of RFID technology into various domains, catalyzed by advancements in radio technology and semiconductor manufacturing. In the 1970s and 1980s, RFID systems began to find utility in supply chain management, enabling real-time tracking of goods and optimizing inventory management processes. This marked the inception of RFID's transformative journey, as businesses leveraged its capabilities to enhance operational efficiency and gain a competitive edge in the marketplace.

The proliferation of RFID technology continued into the 21st century, with its applications diversifying across industries ranging from retail and healthcare to transportation and logistics. In retail environments, RFID-enabled systems revolutionized inventory management practices, offering unprecedented visibility into stock levels and enabling seamless checkout experiences for customers. Similarly, in healthcare settings, RFID tags embedded in medical equipment and patient wristbands enhanced asset tracking and ensured accurate patient identification, thereby enhancing patient safety and operational efficiency.

However, alongside its widespread adoption, RFID technology has also engendered apprehensions regarding security and privacy. The inherent ability of RFID tags to wirelessly transmit data raises concerns about unauthorized access and potential breaches of sensitive information. Moreover, the pervasive nature of RFID-enabled systems has prompted fears of corporate espionage, behavioral profiling, and mass surveillance, prompting calls for stringent regulatory measures and robust security protocols to safeguard individual rights and mitigate risks.

Looking ahead, the future trajectory of RFID technology holds both promise and challenges. As RFID systems become increasingly interconnected and integrated into the fabric of everyday life, addressing security vulnerabilities becomes imperative. Cryptographic techniques, intrusion detection systems, and privacy-enhancing technologies will be pivotal in fortifying RFID systems against emerging threats and ensuring data integrity and confidentiality.

Moreover, interdisciplinary collaboration between stakeholders, including academia, industry, policymakers, and civil society, will be crucial in charting a responsible course for the future development and deployment of RFID technology. By fostering dialogue, promoting transparency, and upholding ethical principles, stakeholders can collectively navigate the complexities of RFID technology and harness its transformative potential for the betterment of society.

In conclusion, this literature review provides a comprehensive analysis of RFID technology, tracing its evolution from its origins to its contemporary applications and the attendant security and privacy concerns. While RFID technology has undeniably revolutionized information systems and catalyzed innovation across industries, its continued advancement necessitates a concerted effort to address security and privacy challenges and ensure responsible stewardship. Through proactive measures and collaborative endeavors, stakeholders can harness the promise of RFID technology while safeguarding individual rights and societal well-being.

PROPOSED METHODOLOGY

The methodology section provides a detailed insight into the approach followed during the development of the RFID and password-based attendance and lock system. It outlines the step-by-step process involved in system design, implementation, and testing, as well as the rationale behind the chosen approach and alternative methodologies considered.

This section begins with an overview of the system design phase, where the architecture, functionalities, and components of the system are conceptualized and planned. It then delves into the implementation phase, where the actual development of the system takes place, encompassing hardware setup, software development, and integration of components. Finally, the testing phase is discussed, focusing on the validation of system functionality, reliability, and performance through various testing methodologies.

The methodology section also provides insight into the rationale behind the chosen approach, emphasizing factors such as scalability, flexibility, and reliability. Alternative methodologies, such as Agile or Waterfall, are briefly considered, highlighting the suitability of the selected approach in balancing project complexity, timeline, and resource constraints.

By following this systematic methodology, the project aims to deliver a robust and reliable RFID and password-based attendance and lock system that meets the needs of modern organizations for enhanced security, efficiency, and user experience.

Now, let us proceed with elaborating on each subheading within the methodology section.

1. System Design:

The system design phase aimed to conceptualize and plan the architecture, functionalities, and components of the RFID and password-based attendance and lock system. The following steps were undertaken:

Requirement Analysis: Comprehensive analysis of user requirements, stakeholders' expectations, and system constraints was conducted to define the scope and objectives of the project.

System Architecture Design: Based on the requirements gathered, the system architecture was designed, outlining the hardware and software components, data flow, and interaction between system modules.

User Interface Design: Prototypes of the user interface for attendance registration, door access control, and system administration were developed to ensure usability and accessibility.

2. Implementation

The implementation phase involved the actual development of the RFID and password-based attendance and lock system based on the design specifications. The following steps were undertaken:

Hardware Setup: The necessary hardware components, including RFID readers, door locks, and the ESP32 DEVKITV1 development board, were assembled and configured according to the system architecture.

Software Development: The system software was developed using appropriate programming languages and frameworks, such as Python for backend development and HTML/CSS/JavaScript for frontend development. Modules for RFID card authentication, password verification, attendance tracking, and cloud integration were implemented.

Integration: The individual software and hardware components were integrated to create a unified system. Compatibility and interoperability between different modules were ensured to achieve seamless operation.

3. Testing

The testing phase aimed to validate the functionality, reliability, and performance of the RFID and password-based attendance and lock system. The following testing methodologies were employed:

Unit Testing: Each module of the system was tested in isolation to verify its functionality and identify any defects or inconsistencies.

Integration Testing: The integrated system was tested to ensure that all components interacted correctly and produced the expected outcomes.

System Testing: End-to-end testing of the entire system was conducted to evaluate its performance in real-world scenarios, including attendance registration, door access control, and cloud synchronization.

User Acceptance Testing: Feedback from users and stakeholders was collected to assess the usability, effectiveness, and satisfaction with the system.

The chosen approach for developing the RFID and password-based attendance and lock system was based on several factors, including the project requirements, available resources, and industry best practices. The rationale for the selected methodology includes:

Scalability: The chosen methodology allows for modular development, enabling future enhancements and scalability to accommodate changing needs.

Flexibility: Iterative development and incremental updates facilitate flexibility in adapting to evolving requirements and addressing emerging challenges.

Reliability: The systematic approach to design, implementation, and testing ensures the reliability and robustness of the system in diverse environments.

HARDWARE AND SOFTWARE REQUIREMENT AND SETUP

Hardware Setup:

I. Microcontroller: Use an Arduino or ESP8266/ESP32 board to control the system.

- Arduino Uno or Arduino Nano for simpler setups.
- ESP8266 (NodeMCU) or ESP32 for Wi-Fi connectivity.

Used in Project: ESP32 DEVKITV1

II. RFID Reader: Choose an RFID reader module compatible with the microcontroller.

- MFRC522 RFID module is commonly used and compatible with Arduino and ESP8266/ESP32.

Used in Project: RFID Reader/Writer RC522 SPI S50 with RFID Card and Tag.

III. Keypad: For password input.

- Matrix keypad or membrane keypad.

Used in Project: 4×4 Matrix Keypad Membrane Switch

IV. Door Lock Mechanism:

- Solenoid lock or servo motor to physically lock/unlock the door.

Used in Project: 12v SOLENOID LOCK

V. Relay:

Used in Project: 12V 1 Channel Relay Module

VI. Display (Optional):

- LCD display for user interface.
- OLED display for compact setups.

Used in Project: 16x2 LCD Display Module

VII. Buzzer (Optional):

- For audio feedback during operations.

Used in Project: 5v Piezoelectric Buzzer

VIII. Power Supply:

- USB power supply or battery pack to power the microcontroller and peripherals.

Used in Project: Standard USB adapter with a microUSB cable (10 W supply)

IX. Wiring:

- Breadboard and jumper wires for prototyping.
- PCB and soldering for a permanent setup.

Used in Project: Jumper Wires and single core copper wire (2mm thickness)

Software Setup:

I. Arduino IDE:

Install the Arduino IDE for programming the microcontroller.

Used in Project: Version 1.8.17

II. Libraries:

Install the necessary libraries for RFID (e.g., MFRC522) and any additional components (e.g., Keypad, LiquidCrystal).

- MFRC522 library for RFID module
- Keypad library for matrix keypad
- LiquidCrystal library for LCD display
-

III. Google Sheets API:

- Set up a Google Sheets API for communication with Google Sheets.
- Create a web application in Google Cloud Console to obtain API credentials.

IV. Programming:

Write the firmware code in Arduino C/C++ to handle RFID card reading, password input, communication with Google Sheets, and control of peripherals.

Now we will explain each section of the setup: -

MICROCONTROLLER: ESP32 DEVKITV1

Brief:

The ESP32 is a highly integrated single-chip solution designed for a wide range of applications including mobile, wearable electronics, and IoT devices. It features a dual-core 32-bit LX6 microprocessor, operating at either 160 or 240 MHz, along with an ultra-low-power co-processor. With its advanced features and low power consumption, the ESP32 is ideal for projects requiring Wi-Fi and Bluetooth connectivity.

Table 4.1: Features of ESP32 DEVKITV1

Feature	Description
Processors	▪ Xtensa dual-core 32-bit LX6 microprocessor operating at 160 or 240 MHz; ▪ Ultra-low power (ULP) co-processor
Memory	▪ 320 KiB RAM ▪ 448 KiB ROM
Wireless connectivity	▪ "Wi-Fi: 802.11 b/g/n ▪ Bluetooth: v4.2 BR/EDR and BLE"
Peripheral interfaces	▪ 34 × programmable GPIOs ▪ 12-bit SAR ADC up to 18 channels ▪ 2 × 8-bit DACs ▪ 10 × touch sensors (capacitive sensing GPIOs) ▪ 4 × SPI ▪ 2 × I²S interfaces ▪ 2 × I²C interfaces ▪ 3 × UART ▪ SD/SDIO/CE-ATA/MMC/eMMC host controller ▪ SDIO/SPI slave controller ▪ Ethernet MAC interface with dedicated DMA and IEEE 1588 Precision Time Protocol support ▪ CAN bus 2.0 ▪ Infrared remote controller (TX/RX up to 8 channels) ▪ Motor PWM ▪ LED PWM (up to 16 channels) ▪ Hall effect sensor ▪ Ultra-low power analog pre-amplifier

Comparison:

The ESP32 stands out as a better option due to the following advantages:

- Integrated Solution: The ESP32 integrates various components including the antenna switch, RF balun, power amplifier, low noise receives amplifier, filters, and power management modules, reducing the number of external components required and minimizing PCB area.

- Advanced Features: With features like fine resolution clock gating, power modes, and dynamic power scaling, the ESP32 offers advanced power management capabilities, making it suitable for low-power applications.

- Versatility: The ESP32 provides a wide range of peripherals and interfaces, including GPIOs, ADC, DAC, SPI, I²S, I²C, UART, Ethernet MAC, CAN bus, PWM, and touch sensors, enabling versatile connectivity and functionality in IoT projects.

- Wireless Connectivity: Supporting Wi-Fi 802.11 b/g/n and Bluetooth v4.2 BR/EDR and BLE, the ESP32 offers robust wireless connectivity for data transfer and communication.

- High Performance: Equipped with a dual-core Xtensa LX6 microprocessor, operating at up to 240 MHz, and a dedicated ultra-low-power co-processor, the ESP32 delivers high performance and efficiency for demanding applications.

- Overall, the ESP32's integration, advanced features, versatility, wireless capabilities, and performance make it a superior choice for a wide range of IoT and embedded projects compared to other alternatives.

Overall, the ESP32's integration, advanced features, versatility, wireless capabilities, and performance make it a superior choice for a wide range of IoT and embedded projects compared to other alternatives.

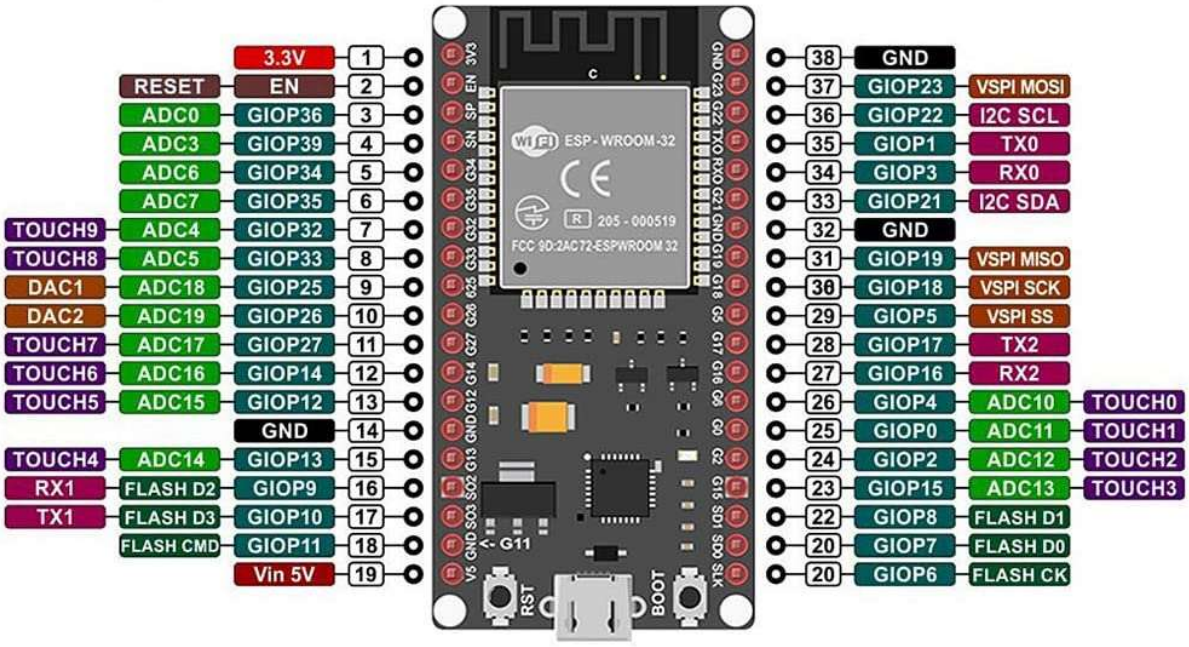


Fig 4.1: ESP32 DEVKITV1 Pinout

RFID Module: MFRC522 Reader/Writer

Brief:

The MFRC522 is a highly integrated reader/writer IC for contactless communication at 13.56 MHz. The MFRC522 reader supports ISO/IEC 14443 A/MIFARE and NTAG. The MFRC522's internal transmitter can drive a reader/writer antenna designed to communicate with ISO/IEC 14443 A/MIFARE cards and transponders without additional active circuitry. The receiver module provides a robust and efficient implementation for demodulating and decoding signals from ISO/IEC 14443 A/MIFARE compatible cards and transponders. The digital module manages the complete ISO/IEC 14443 A framing and error detection (parity and CRC) functionality.

The MFRC522 supports MF1xxS20, MF1xxS70 and MF1xxS50 products. The MFRC522 supports contactless communication and uses MIFARE higher transfer speeds up to 848 kBd in both directions.

The following host interfaces are provided:

• Serial Peripheral Interface (SPI)

• Serial UART (like RS232 with voltage levels dependant on pin voltage supply)

• I2C-bus interface

Features: -

- Highly integrated analog circuitry to demodulate and decode responses

- Buffered output drivers for connecting an antenna with the minimum number of external components
- Supports ISO/IEC 14443 A/MIFARE and NTAG
- Typical operating distance in Read/Write mode up to 50 mm depending on the antenna size and tuning
- Supports MF1xxS20, MF1xxS70 and MF1xxS50 encryption in Read/Write mode
- Supports ISO/IEC 14443 A higher transfer speed communication up to 848 kBd
- Supports MFIN/MFOUT
- Additional internal power supply to the smart card IC connected via MFIN/MFOUT
- Supported host interfaces
    - SPI up to 10 Mbit/s
    - I2C-bus interface up to 400 kBd in Fast mode, up to 3400 kBd in High-speed mode
    - RS232 Serial UART up to 1228.8 kBd, with voltage levels dependant on pin voltage supply
- FIFO buffer handles 64 byte send and receive
- Flexible interrupt modes
- Hard reset with low power function
- Power-down by software mode
- Programmable timer
- Internal oscillator for connection to 27.12 MHz quartz crystal
- 2.5 V to 3.3 V power supply
- CRC coprocessor
- Programmable I/O pins
- Internal self-test

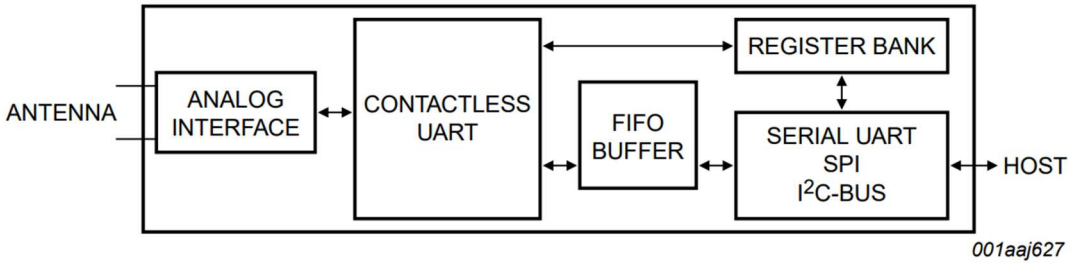


Fig 4.2: Simplified Block Diagram of the MFRC522

The analog interface handles the modulation and demodulation of the analog signals.

The contactless UART manages the protocol requirements for the communication protocols in cooperation with the host. The FIFO buffer ensures fast and convenient data transfer to and from the host and the contactless UART and vice versa.

Various host interfaces are implemented to meet different customer requirements.

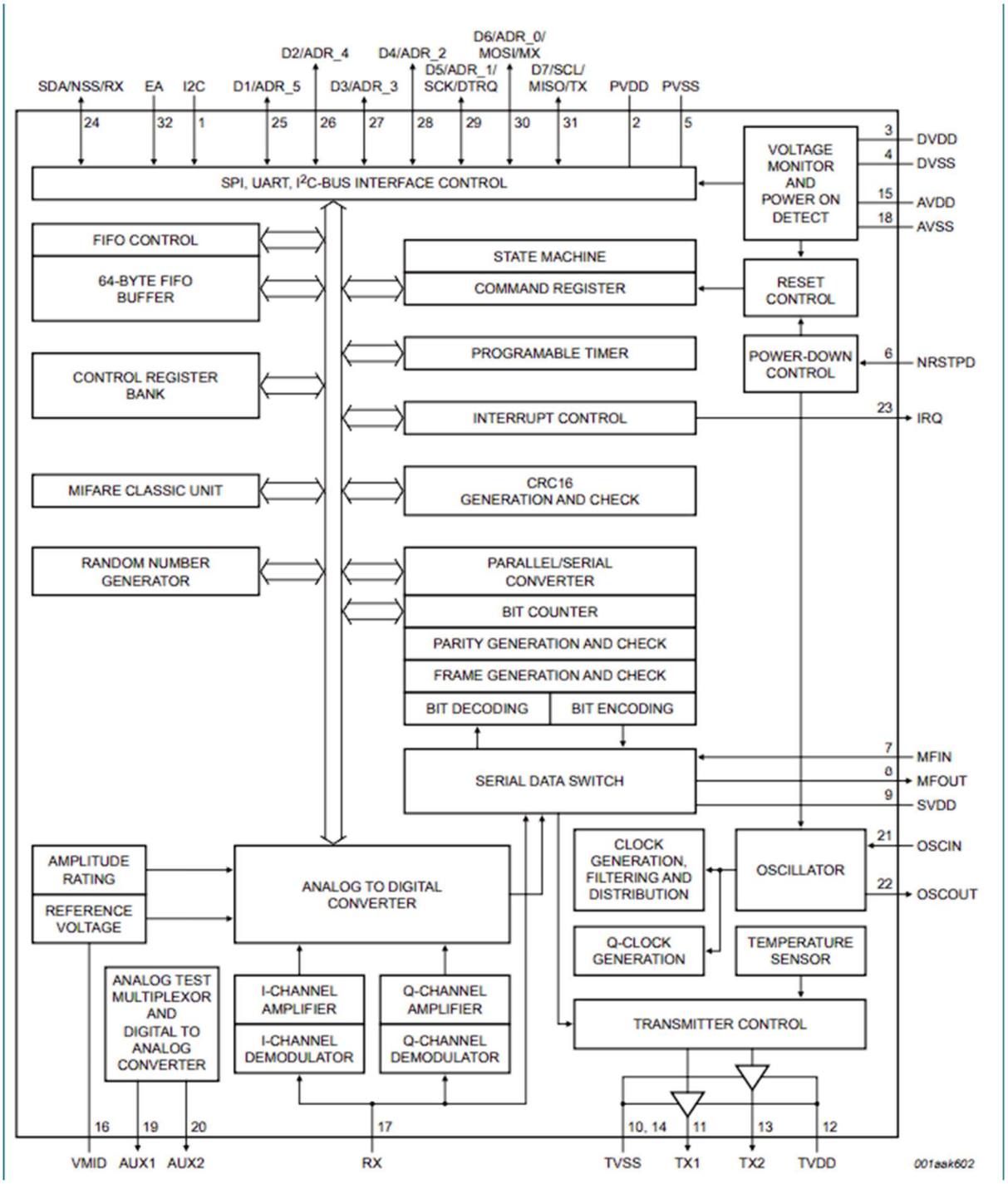


Fig 4.3: Detailed block diagram of MFRC522

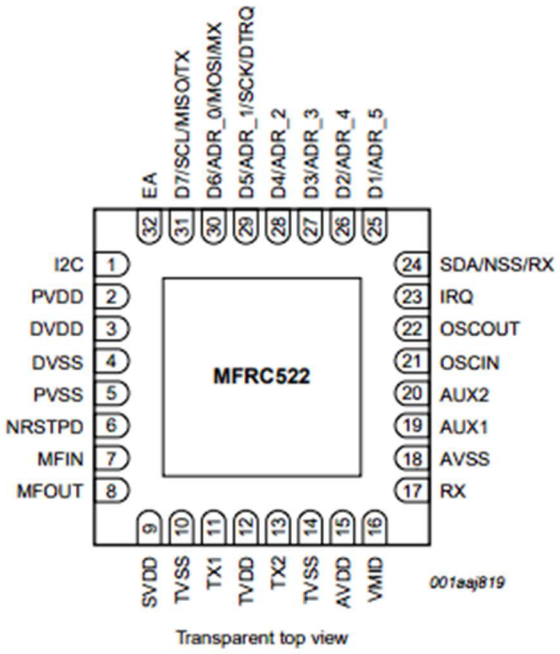


Fig 4.4: Pinning configuration HVQFN32 (SOT617-1)

Functional Description

The MFRC522 transmission module supports the Read/Write mode for ISO/IEC 14443 A/MIFARE using various transfer speeds and modulation protocols.

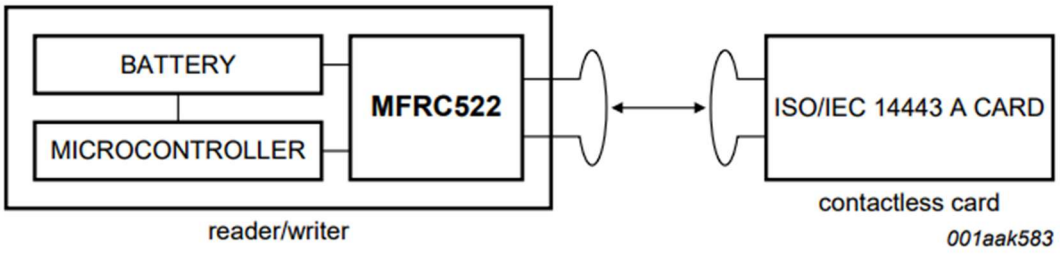


Fig 4.5: MFRC522 Read/Write mode



Fig 4.6: RFID MFRC522 MODULE

KEYPAD 4x4 Membrane Switch

This 16-button keypad provides a useful human interface component for microcontroller projects. Convenient adhesive backing provides a simple way to mount the keypad in a variety of applications. The Keypad 4x4 features a total of 16 buttons in Matrix form. This is a membrane keypad with no moving parts. It has a nice overlay depicting a telephone type keypad with additional four functional buttons. A female 8-pin berg connector is provided for interfacing it with your microcontroller circuits.

Features

- Ultra-thin design
- Adhesive backing
- Excellent price/performance ratio
- Easy interface to any microcontroller

Application Ideas

- Security systems
- Menu selection
- Data entry for embedded systems

Key Specifications

- Maximum Rating: 24 VDC, 30 mA
- Interface: 8-pin access to 4x4 matrix
- Operating temperature: 32 to 122 °F (0 to 50°C)
- Dimensions: Keypad, 2.7 x 3.0 in (6.9 x 7.6 cm)

Specifications

- Cable Length: 3-1/3" or 85mm (include connector)
- Weight: 8g
- Connector: Dupont 8 pins, 0.1" (2.54mm) Pitch
- Mount Style: Self-Adherence
- Max. Circuit Rating: 24VDC, 30mA
- Insulation Spec.: 100M Ohm, 100V
- Dielectric Withstand: 250VRms (60Hz, 1min)
- Contact Bounce: <=5ms
- Life Expectancy: 1 million closures
- Operation Temperature: -20 to +40 °C



Fig 4.7: Keypad 4x4 Membrane Module

How it Works

Matrix keypads use a combination of four rows and four columns to provide button states to the host device, typically a microcontroller. Underneath each key is a pushbutton, with one end connected to one row, and the other end connected to one column.

These connections are shown below.

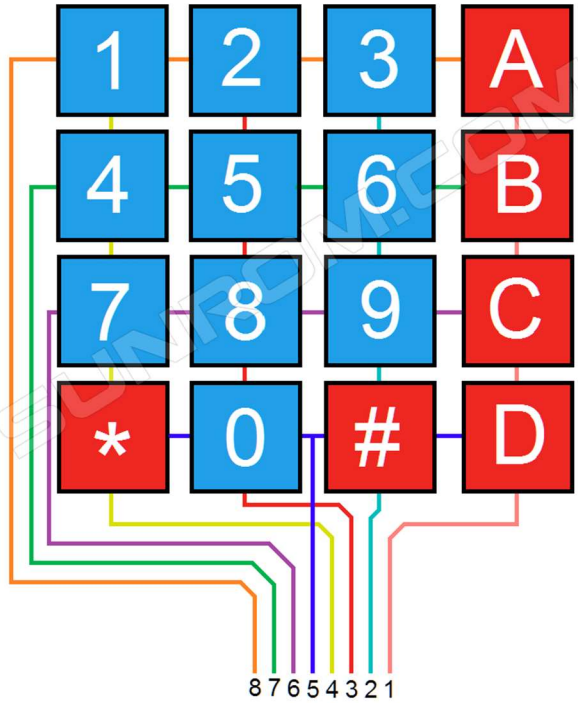


Fig 4.8: Pinning Diagram of Keypad

SOLENOID LOCK

Features

- High quality ultra-compact electric lock.
- Rustproof, durable, safe, convenient to use.
- Suction which tightly sucks the iron, thus locking the door.
- Applicable for being installed in the escape door or fire door electronic controlled system.
- Adopts the principle of electric magnetism, when the current through the silicon, the electromagnetic lock will achieve a strong.

Specifications

- Body Material: Iron
- Rated voltage: DC 12V.
- Rated current: 0.80A.
- Power: 9.6 Watt
- Holding force: 0.25kg
- Energized forms: intermittent
- Unlocking time: 1S
- Weight: 150 gm
- Dimensions: 54 x 42 x 28 mm (LxWxH)
- Cable length: 20 cm

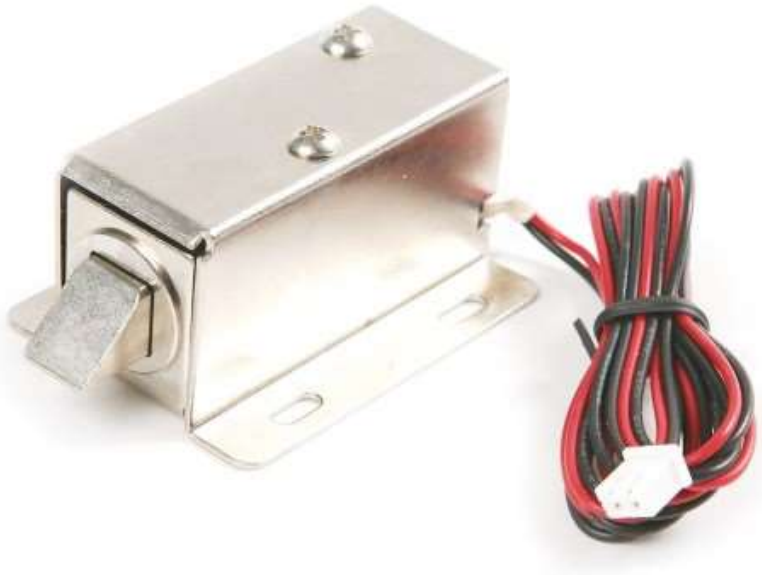


Fig 4.9: Solenoid Lock

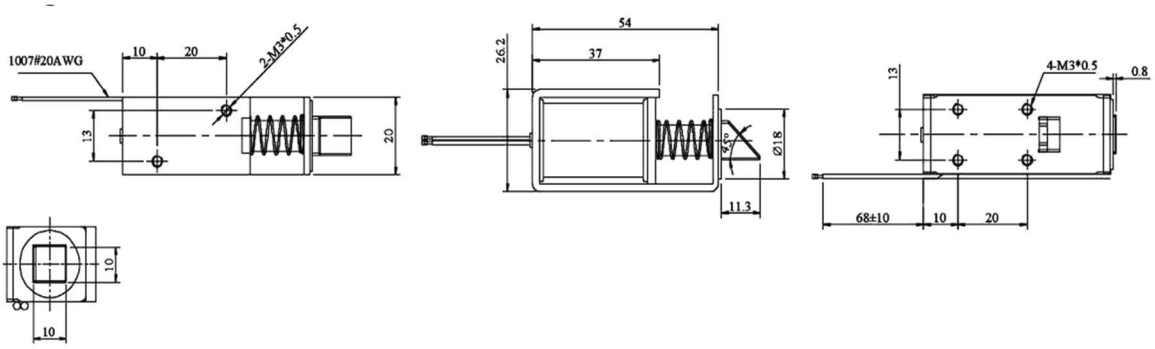


Fig 4.10: Dimensions of Solenoid Lock

16x2 LCD Display

A 16x2 LCD display is a common type of alphanumeric display that consists of 16 columns and 2 rows, hence the name "16x2". Each character cell in the display can display a single alphanumeric character or symbol. These displays are widely used in various electronic projects and devices due to their simplicity, low cost, and ease of use.

Here are some key features of a typical 16x2 LCD display:

- Size: The display typically measures around 80mm x 36mm.
- Resolution: 16 characters per line and 2 lines.
- Interface: It usually uses a parallel interface for communication with microcontrollers or other devices. Some models also support serial interfaces like I2C or SPI.
- Backlight: Many 16x2 LCD displays come with a built-in LED backlight for improved visibility in low-light conditions.
- Character Set: Supports a standard ASCII character set, allowing the display of letters, numbers, punctuation marks, and some special symbols.
- Controller: Most 16x2 LCD displays are based on the HD44780 or compatible controller chip, which simplifies interfacing with microcontrollers.

These displays are commonly used in various applications such as digital clocks, thermometers, electronic meters, DIY projects, and appliances. They are relatively easy to use with microcontrollers like Arduino, Raspberry Pi, and others, making them a popular choice for hobbyists and electronics enthusiasts.



Fig 4.11: LCD Module with Proper Dimensions

Pinout and Pin description

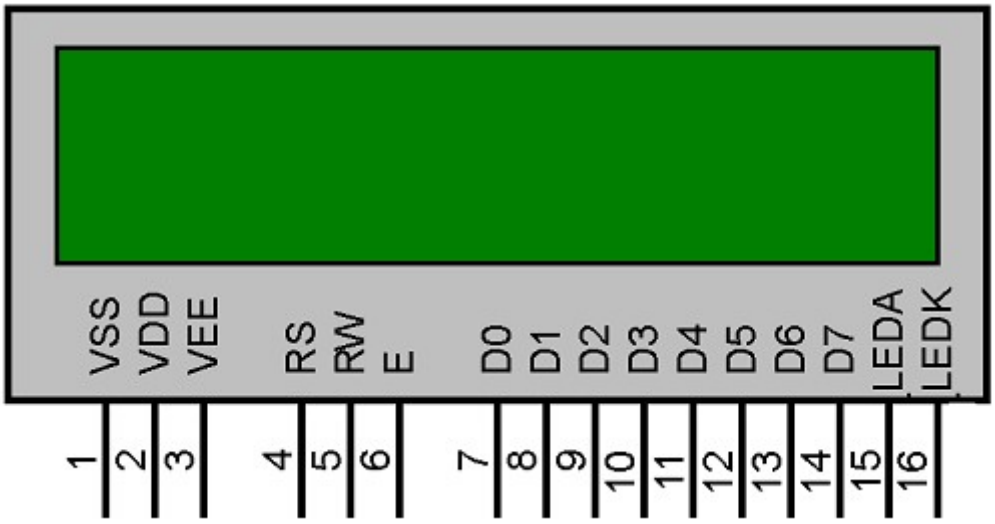


Fig 4.12: 16x2 LCD Pinout

Table 4.2: Pin Description

Sr. No.	Pin No.	Pin Name	Pin Type	Pin Description	Pin Connection
1	Pin 1	Ground	Source Pin	This is a ground pin of LCD	Connected to the ground of the MCU/Power source
2	Pin 2	VCC	Source Pin	This is the supply voltage pin of LCD	Connected to the supply pin of Power source
3	Pin 3	V0/VEE	Control Pin	Adjusts the contrast of the LCD	Connected to a variable POT that can source 0-5V
4	Pin 4	Register Select	Control Pin	Toggles between Command/Data Register	Connected to an MCU pin and gets either 0 or 1. 0-> Command Mode 1-> Data Mode
5	Pin 5	Read/Write	Control Pin	Toggles the LCD between	Connected to an MCU

				Read/Write Operation	pin and gets either 0 or 1. 0 -> Write Operation 1-> Read Operation
6	Pin 6	Enable	Control Pin	Must be held high to perform Read/Write Operation	Connected to MCU and always held high
7-14	Pin 7-14	Data Bits (0-7)	Data/Command Pin	Pins used to send Command or data to the LCD.	In 4-Wire Mode Only 4 pins (0-3) are connected to MCU. In 8-Wire Mode All 8 pins(0-7) are connected to MCU
8	Pin 15	LED Positive	LED Pin	Normal LED-like operation to illuminate the LCD	Connected to +5V
9	Pin 16	LED Negative	LED Pin	Normal LED-like operation to illuminate the LCD connected with GND	Connected to ground

Buzzer

A piezoelectric buzzer is a type of electronic audio signaling device that produces sound when an electrical signal is applied to it. It utilizes the piezoelectric effect, which occurs in certain materials when they are subjected to mechanical stress, such as pressure or vibration, resulting in the generation of an electric charge.

Specifications:

- Input Voltage(Max.) : 5V
- Resistance: 90 Ω
- Resonance Frequency: 2048 Hz
- Sound pressure(dB(A)/10cm)min.: 80
- Body Size : 12 x 9.5mm
- Pin Pitch: 6mm
- External Material: Plastic;
- Color: Black

Table 4.3: Buzzer Specifications

Attribute	Value
Frequency (Hz)	2048
Pin Spacing (mm)	6
Material	Plastic
Color	Black
Width (mm)	12 (Body Diameter)
Height (mm)	9.5
Weight (g)	1 (approx) (each)
Weight	0.015 kg
Dimensions	8 × 10 × 1 cm

Relay Module

12V 1 Channel Relay Module is a small and easy to use. Use it to control one 240V power appliance directly from Arduino, Raspberry Pi, and other microcontrollers or low voltage circuits. Perfect for switching 240V appliances – lights, fans, etc. and even high-power motors at lower voltages.

The board uses a high-quality relay, which can handle a maximum of 10A/250 V AC or 15A/125V AC. Each relay has all three connections – Common, Normally Open, Normally Closed brought out to 3 pin screw terminals which make it easy to make and remove connections. The board has a power indication and a relay status LED to ease debugging. The board can accept control inputs within a wide range of voltages from 4V to 12V.

Power input and relay control signals are brought to header pins on the board. Hence, the board can be easily interfaced with our development boards using our female-to-female jumper wires.

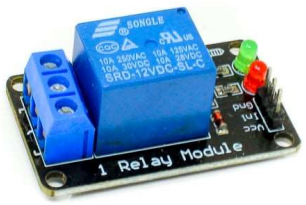


Fig 4.12: Relay Module

Relay modules 1-channel features
• Contact current 10A and 250V AC or 30V DC.
• Each channel has indication LED.
• Coil voltage 12V per channel.
• Kit operating voltage 5-12 V
• Input signal 3-5 V for each channel.
• Three pins for normally open and closed for each channel.

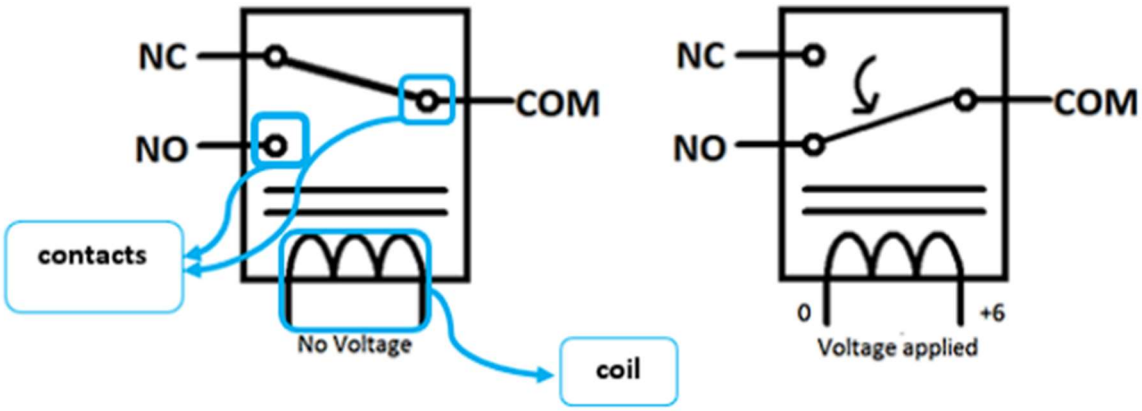


Fig 4.13: Working Principle of Relay Module

Main Setup

In the main setup section, we initialize and configure the various components of our project to ensure they function correctly during runtime. This section typically includes initializing communication interfaces, setting pin modes, and configuring any necessary parameters for sensor or display modules. It is crucial to establish proper communication channels between the microcontroller (ESP32) and peripherals to enable seamless data exchange and interaction. Additionally, this section may involve setting up interrupt service routines, initializing libraries, and performing any pre-runtime calibration or initialization steps required for optimal performance.

During this setup phase, we establish the groundwork for the project's functionality, laying the foundation for subsequent operations in the main loop. By carefully configuring each component and ensuring proper initialization, we create a stable environment for the execution of the project's core logic. This setup phase is critical for ensuring the reliability, accuracy, and efficiency of our system as it transitions from initialization to active operation. Therefore, thorough testing and validation of the setup parameters are essential to identify and rectify any potential issues before proceeding to the main execution phase.

Table 4.4: Pin Configurations

Pin on ESP32	Pin Designation	Component
D5	RST	MFRC522
D21	SS	MFRC522
D23	MOSI	MFRC522
D19	MISO	MFRC522
D18	SCK	MFRC522
3V3	3.3V	MFRC522
GND	GND	MFRC522
D32	rowPins	KEYPAD_4x4
D33	rowPins	KEYPAD_4x4
D25	rowPins	KEYPAD_4x4
D26	rowPins	KEYPAD_4x4
D27	colPins	KEYPAD_4x4
D14	colPins	KEYPAD_4x4
D12	colPins	KEYPAD_4x4
D13	colPins	KEYPAD_4x4
Vin	VCC	LCD_16x2
GND	GND	LCD_16x2
D21	SDA	LCD_16x2
D22	SCL	LCD_16x2

MFRC522 RFID Module Setup:

Connect the MFRC522 RFID module to the ESP32 as follows:

- RST pin to D5
- SS pin to D21
- MOSI pin to D23
- MISO pin to D19
- SCK pin to D18
- 3.3V pin to 3V3
- GND pin to GND

Keypad 4x4 Setup:

Connect the keypad to the ESP32 as follows:

- Row pins to D32, D33, D25, and D26
- Column pins to D27, D14, D12, and D13

LCD 16x2 Setup:

Connect the LCD 16x2 to the ESP32 as follows:

- VCC pin to Vin
- GND pin to GND
- SDA pin to D21
- SCL pin to D22

Main Setup:

- Initialize the MFRC522 module for RFID functionality.
- Initialize the keypad for input.
- Initialize the LCD for output.
- Set up any additional functionality required for your project, such as user authentication via RFID or keypad input, and display feedback on the LCD.

Circuit Diagram

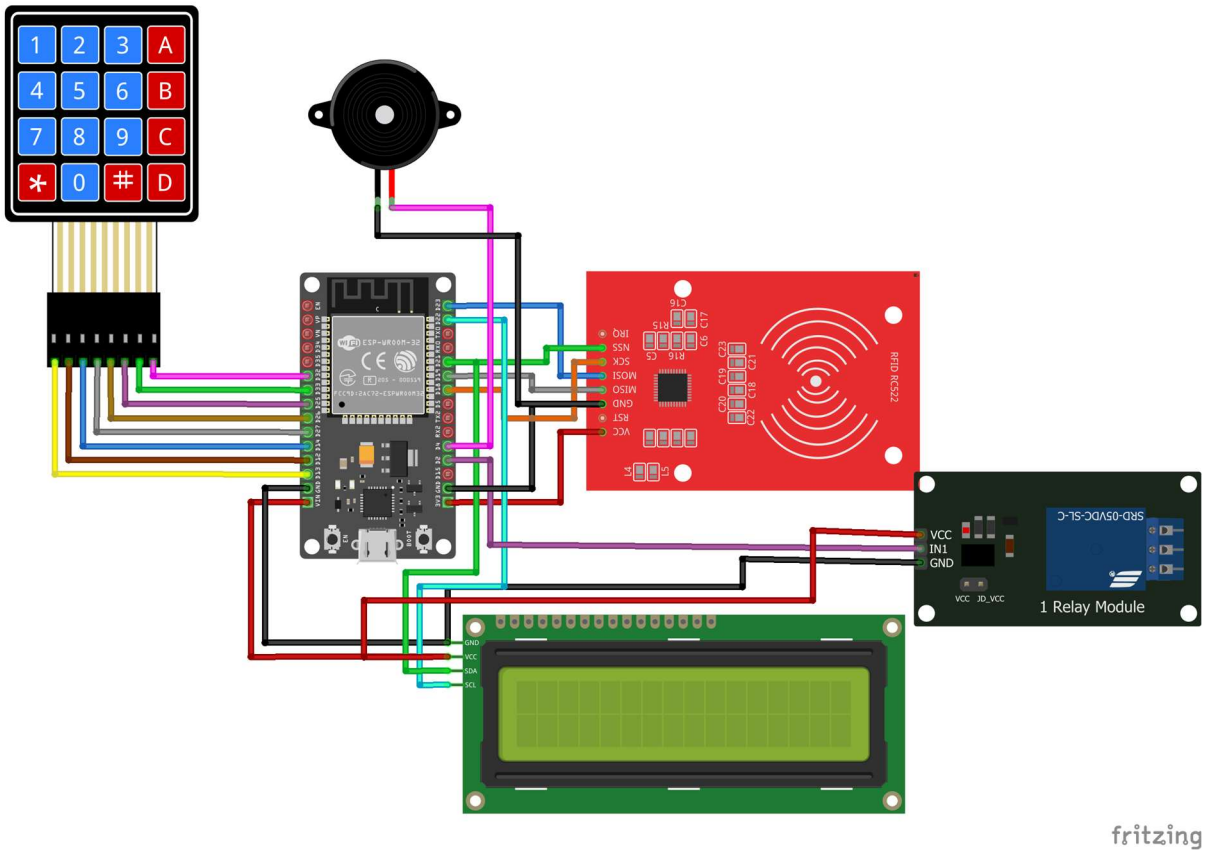


Fig 4.14: Circuit Diagram

Algorithm

Algorithm for RFID Card Authentication System with WiFi Integration

1. Initialization:
   - Initialize serial communication to communicate with the Arduino board.
   - Initialize the Liquid Crystal Display (LCD) for user interface.
   - Establish a connection to the local WiFi network for internet access.
   - Initialize the Serial Peripheral Interface (SPI) bus for communication with the RFID module.
   - Set pin modes for the relay and buzzer for controlling physical access.
   - Initialize Preferences object for managing stored passwords.
   - Set default password to "0000" if no password is stored in the EEPROM.

2. Main Loop:
   - Display "Tap or press *" on the LCD to prompt user action.
   - Check for the presence of a new RFID card:
     - If no card is detected:
       - Check for input from the keypad:
         - If input is "*", prompt the user to enter or change the password.
     - If a card is detected:

- Read the Unique Identifier (UID) of the card and attempt to read data from a specific block.
- Convert the UID to a string and display it on the LCD.
- Verify the authorization status of the card by querying a Google Sheets script with the UID.
- If authorized:
  - Display "Registered" on the LCD, activate a brief buzzer sound, open the relay for physical access, send data to Google Sheets, and close the relay.
- If unauthorized:
  - Display "Unauthorised" on the LCD, activate a longer buzzer sound, and take no further action.

3. WiFi Connectivity:
- Attempt to connect to the predefined WiFi network.
- Display "Online" or "Offline" on the LCD based on the connection status.
- Retry connection if unsuccessful.

4. Functions:
-sendDataToSheet()`:
- Construct a URL with the card holder's name and send an HTTP GET request to a Google Sheets script.
- Display "Confirmed" on the LCD if successful.

-checkifAuthorised()`:
- Construct a URL with the card UID and send an HTTP GET request to a Google Sheets script.
- Parse the response ("true" or "false") and store it for authorization.

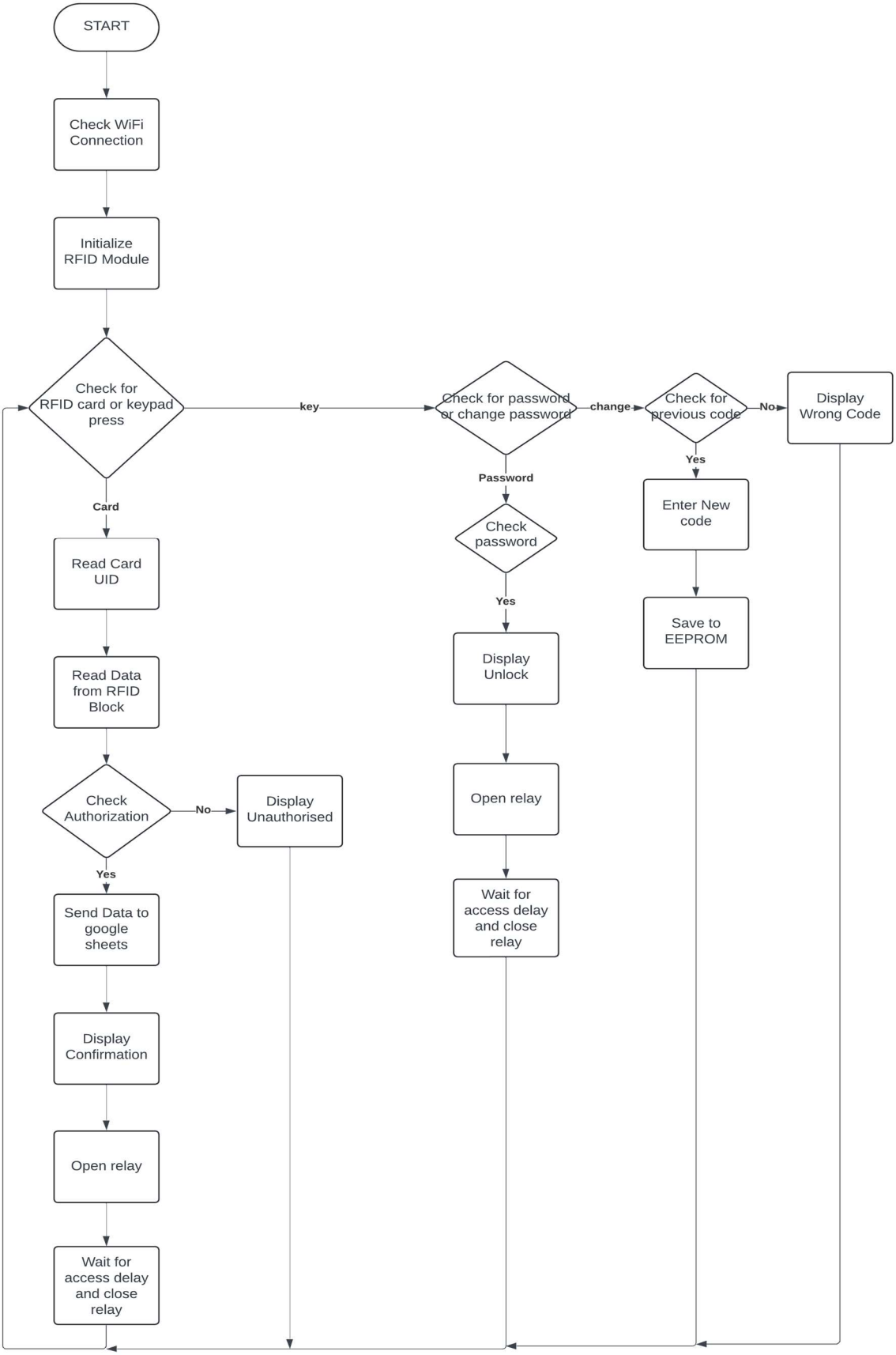
-ReadDataFromBlock(int blockNum, byte readBlockData[])`:
- Authenticate the RFID card using a default key.
- Read data from a specific block of the RFID card.

-getPassword()`:
- Prompt the user to enter or change the password using the keypad.
- Store the new password if it matches the previous password.
- If the entered password matches the stored password, activate a brief buzzer sound and open the relay.
- If the entered password is incorrect, activate a longer buzzer sound.

5. Helper Functions:
-initializePreferencesWithDefaultPassword()`:
- Set a default password ("0000") if no password is stored in the EEPROM.

This algorithm provides a structured overview of the RFID card authentication system's functionality, including initialization, main loop operations, individual functions, and helper functions.

Flowchart


START
Check WiFi Connection
Initialize RFID Module
Check for RFID card or keypad press
key
Check for password or change password
change
Check for previous code
No
Display Wrong Code
Yes
Enter New code
Save to EEPROM
Password
Check password
Yes
Display Unlock
Open relay
Wait for access delay and close relay
Card
Read Card UID
Read Data from RFID Block
Check Authorization
No
Display Unauthorised
Yes
Send Data to google sheets
Display Confirmation
Open relay
Wait for access delay and close relay

Source Code

```
// Module 1: RFID Authentication Module
class RFIDAuthentication {
  public:
    void initializeModule();
    void readCardUID(String &cardUID);
    void readDataFromBlock(int blockNum, byte readBlockData[], byte &bufferLen);
};

// Module 2: Keypad Input Module
class KeypadInput {
  public:
    char getKey();
};

// Module 3: WiFi Connectivity Module
class WiFiConnectivity {
  public:
    void connectToWiFi();
};

// Module 4: HTTP Requests Module
class HTTPRequests {
  public:
    String sendGETRequest(String url);
};

// Module 5: EEPROM Preferences Module
class EEPROMPreferences {
  public:
    void initializePreferences();
    String getPassword();
    void setPassword(String password);
};

// Module 6: LCD Display Module
class LCDDisplay {
  public:
    void initializeDisplay();
    void showMessage(String message);
};

// Module 7: Access Control Module
class AccessControl {
  public:
    void initializePins();
    void activateRelay();
    void activateBuzzer();
};
```

```
// Module 8: Main Code Module
class MainCode {
  private:
    RFIDAuthentication rfidAuth;
    KeypadInput keypadInput;
    WiFiConnectivity wifiConnectivity;
    HTTPRequests httpRequests;
    EEPROMPreferences eepromPreferences;
    LCDDisplay lcdDisplay;
    AccessControl accessControl;

  public:
    void setup();
    void loop();
};
```

Module 1: RFID Authentication Module

Functionality: This module is responsible for detecting RFID cards, reading their Unique Identifier (UID), and verifying the data for authentication purposes. It ensures that only authorized RFID cards are granted access to the system.

Module 2: Keypad Input Module

Functionality: This module handles user input from a keypad, primarily used for password entry and management. It facilitates the interaction between users and the system for password-based authentication and password change requests.

Module 3: WiFi Connectivity Module

Functionality: This module manages the connection to the local WiFi network, allowing the system to communicate with external servers or services over the internet. It ensures seamless and reliable internet connectivity for data transmission.

Module 4: HTTP Requests Module

Functionality: The HTTP Requests Module is responsible for sending HTTP GET requests to a Google Sheets script for authentication and data logging purposes. It enables communication between the system and external web services to exchange data securely.

Module 5: EEPROM Preferences Module

Functionality: This module is tasked with managing the storage and retrieval of passwords from EEPROM (Electrically Erasable Programmable Read-Only Memory). It provides a persistent storage solution for storing sensitive data such as passwords securely.

Module 6: LCD Display Module

Functionality: The LCD Display Module provides a user interface through an I2C LCD display. It enables the system to communicate information, status updates, and prompts to users in a clear and understandable manner via the LCD screen.

Module 7: Access Control Module

Functionality: This module controls physical access to the system through the activation of a relay and buzzer. It manages the mechanism for granting or denying access based on the authentication results, ensuring the security of the system.

Module 8: Main Code Module

Functionality: The Main Code Module serves as the central orchestrator of the system's main logic. It coordinates the interactions between all other modules, handling the flow of operations, and executing the core functionality of the system.

These modules work together seamlessly to create a robust and efficient access control system, ensuring proper authentication and authorization while providing a user-friendly experience.

Results

RFID Authentication

- The system successfully reads RFID cards placed near the reader.
- It accurately extracts the unique identification (UID) of each RFID card for further processing.

Keypad Input

- The system effectively captures input from the keypad for password entry and management.
- It provides a user-friendly interface for users to interact with the system through the keypad.

WiFi Connectivity

- The system seamlessly connects to the local WiFi network for data transmission and communication.
- It ensures stable and reliable connectivity for accessing remote services.

HTTP Requests

- The system sends HTTP GET requests to the Google Sheets script for authentication and data logging.
- It retrieves responses from the server to determine card authorization status and update attendance records.

EEPROM Preferences

- The system securely stores and retrieves passwords from EEPROM for access control purposes.
- It maintains password integrity across system reboots and power cycles.

LCD Display

- The system utilizes an I2C LCD display to provide clear and concise feedback to users.
- It displays relevant information such as authentication status and system prompts for user interaction.

Access Control

- The system controls physical access through the relay and alerts users with the buzzer.
- It grants access upon successful authentication and denies access to unauthorized users.

Overall System Performance

- The integrated system functions reliably and efficiently in real-world scenarios.
- It provides robust RFID-based access control with additional security features such as password authentication.
- The system offers a user-friendly interface and seamless connectivity for enhanced user experience.

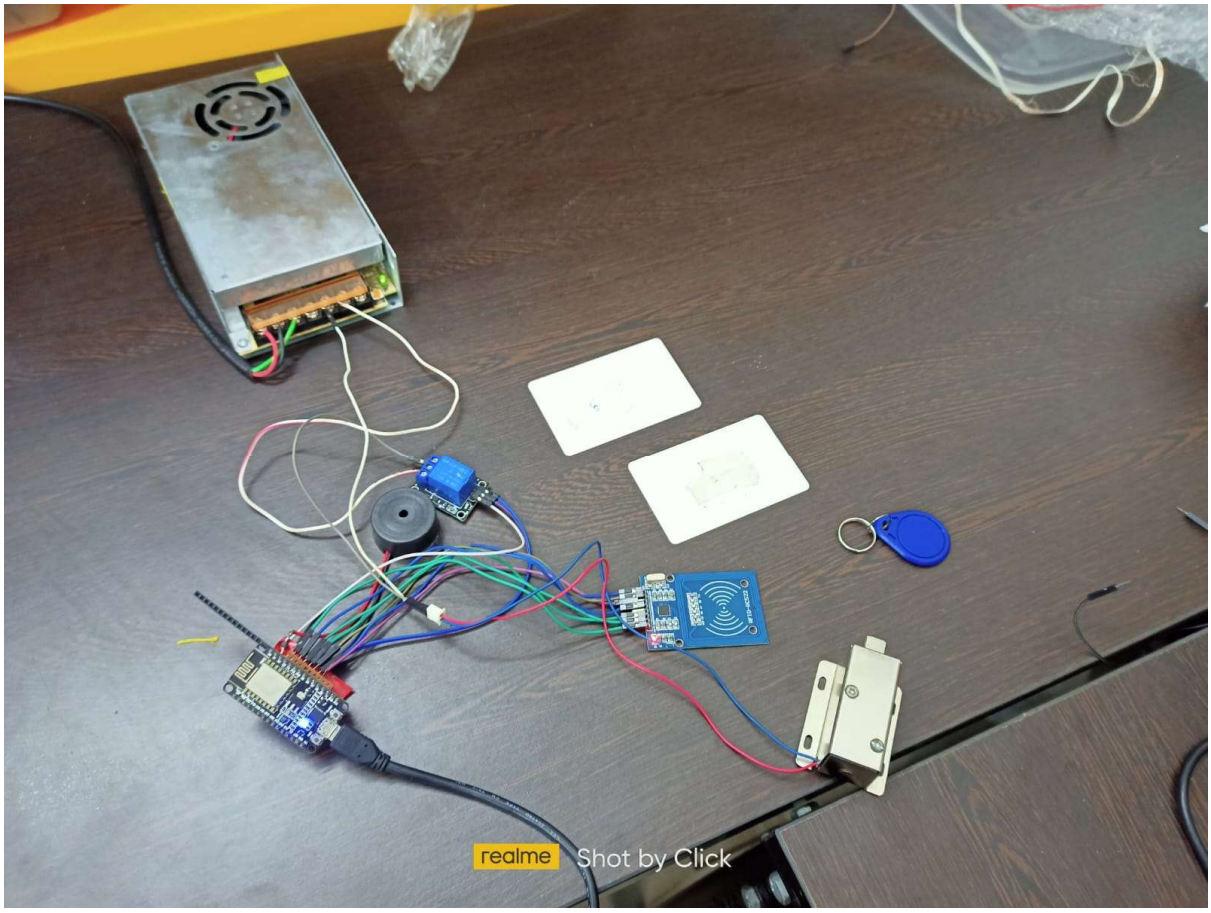
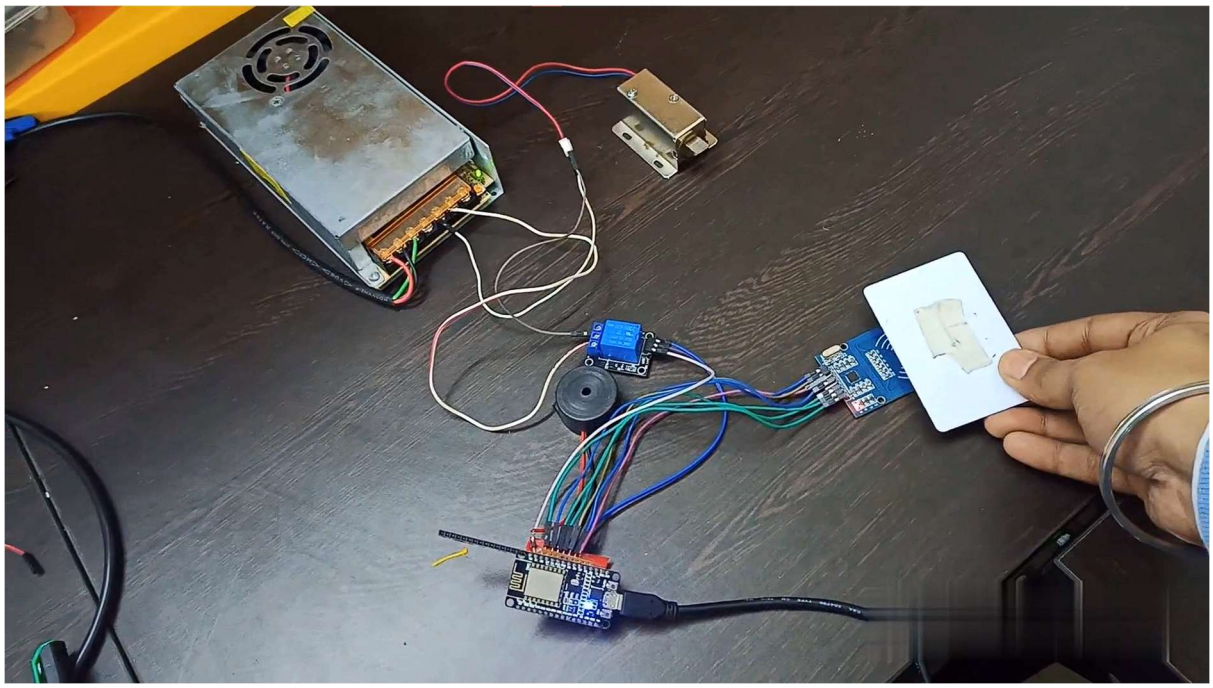


Fig 5.1: Initial Setup

Now we will test a authorize card to the system


Fig 5.2: Card Scanning in RFID

After verifying the lock is open.

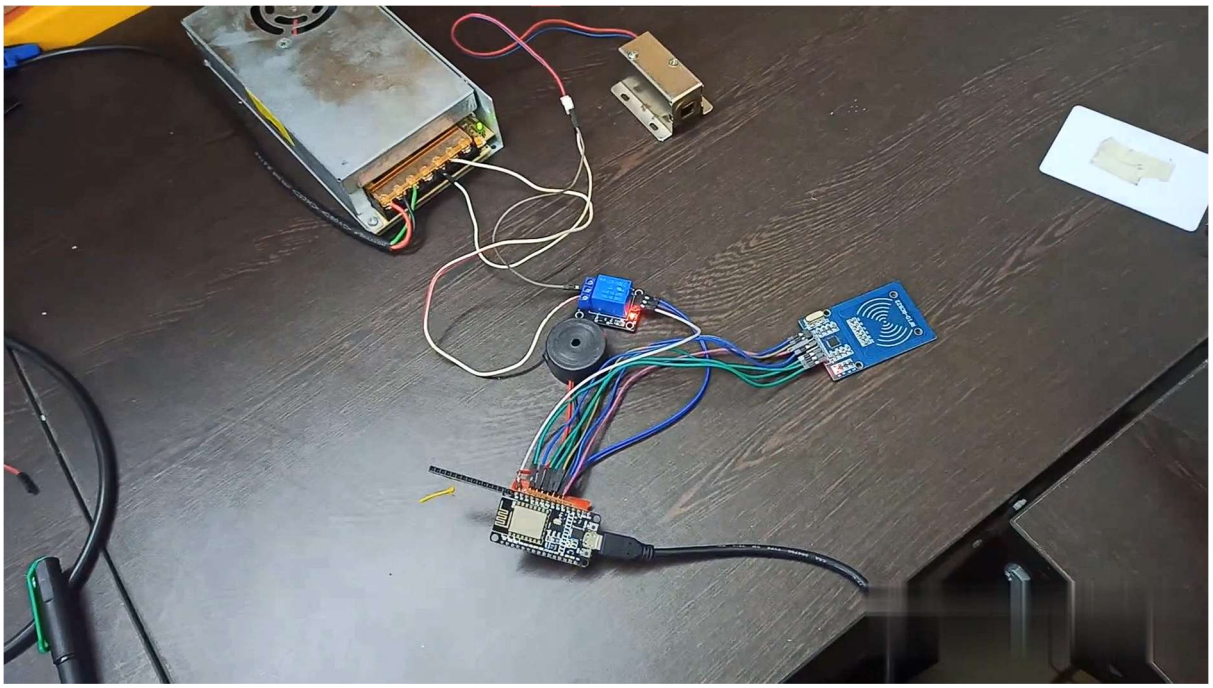


Fig 5.3: Lock is open after verifying

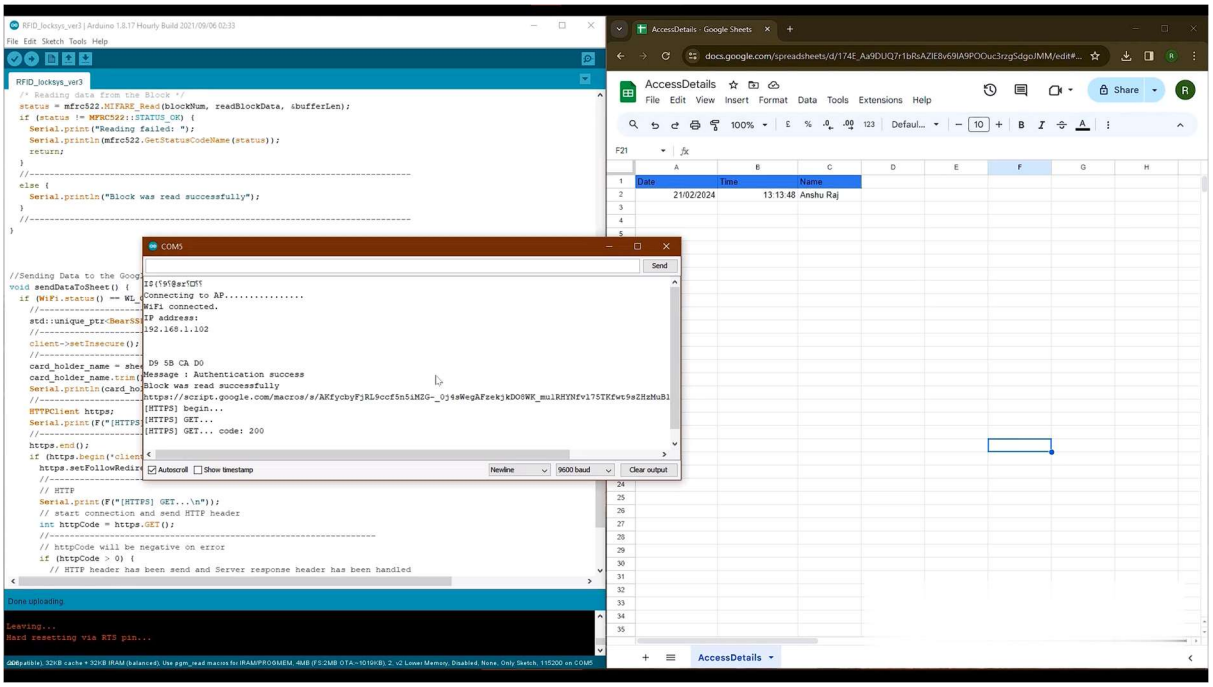


Fig 5.4: Software part of above results

The authorization and attendance registration components of the system seamlessly integrate RFID card authentication with remote server communication to provide a comprehensive access control solution. Upon presenting an RFID card, the system verifies its authorization status by querying a Google Sheets script, determining whether the card is registered in the system. Upon successful authorization, the system registers the attendance by sending relevant data to the Google Sheets script, facilitating real-time attendance tracking. This

automated process ensures accurate and efficient attendance management while maintaining robust security measures. Through effective utilization of RFID technology and remote server communication, the system streamlines access control operations and enhances overall organizational efficiency.

Table 5.1: Performance Test Data

Card ID	Card Status	Lock System	Attendance	Status	Overall Status	Remarks
D95BCAD0	Authorized	Unlocked	Present	Authorized	Success	
4AD1E234	Authorized	Unlocked	Present	Authorized	Success	
A14AD283	Authorized	Unlocked	Didn't Send	Authorized	Failure	Problem with https GET request
F8E1C6B2	Unauthorized	Locked	Nil	Unauthorized	Success	
6B5A8F7C	Unauthorized	Locked	Nil	Unauthorized	Success	

Sometimes, there might be instances where the HTTPS GET request encounters errors. These errors could occur due to various reasons such as network connectivity issues, server unavailability, or invalid URLs. It's essential to handle these errors gracefully to ensure the reliability and robustness of the system. By incorporating error handling mechanisms, such as retry strategies, logging, and alert notifications, we can effectively manage and mitigate these errors. Additionally, thorough testing and monitoring of the system can help identify and address any potential issues related to HTTPS GET requests, ensuring smoother operation and improved performance overall.

Table 5.2: Average Card Read Distance

Card ID	Card Status	RFID Read Status	Distance (cm)
D95BCAD0	Authorized	Success	3.5
4AD1E234	Authorized	Success	3.3
A14AD283	Authorized	Success	2.3
F8E1C6B2	Unauthorized	Success	2.7
6B5A8F7C	Unauthorized	Success	3.3

Table 5.2 presents the average card read distance for both authorized and unauthorized cards. The average distance for authorized cards is calculated to be approximately 3.03 centimeters, while unauthorized cards have an average read distance of approximately 3 centimeters. This data suggests that there is a slight variation in the average read distance between authorized and unauthorized cards, with authorized cards having a marginally higher average read distance. These averages provide insights into the typical operating range of the RFID reader system and can be used to optimize the placement and configuration of the reader for efficient card detection and authentication.



Fig 5.5: Final Setup with password integration

Table 5.3: Performance Test Data for password

Scenario	Status for Lock	System Status
Password Unlock	Unlock	Success
Password Change Requested (Wrong Password)	Locked	Success
Password Change Requested (Right Password)	Locked	Success
Password Wrong	Locked	Success

The authorization and attendance registration components of the system seamlessly integrate RFID card authentication with remote server communication to provide a comprehensive access control solution. Upon presenting an RFID card, the system verifies its authorization status by querying a Google Sheets script, determining whether the card is registered in the system. Upon successful authorization, the system registers the attendance by sending relevant data to the Google Sheets script, facilitating real-time attendance tracking. This automated process ensures accurate and efficient attendance management while maintaining robust security measures. Through effective utilization of RFID technology and remote server communication, the system streamlines access control operations and enhances overall organizational efficiency.

Sometimes, there might be instances where the HTTPS GET request encounters errors. These errors could occur due to various reasons such as network connectivity issues, server unavailability, or invalid URLs. It's essential to handle these errors gracefully to ensure the reliability and robustness of the system. By incorporating error handling mechanisms, such as retry strategies, logging, and alert notifications, we can effectively manage and mitigate these errors. Additionally, thorough testing and monitoring of the system can help identify and

address any potential issues related to HTTPS GET requests, ensuring smoother operation and improved performance overall.

Table 5.2 presents the average card read distance for both authorized and unauthorized cards. The average distance for authorized cards is calculated to be approximately 3.03 centimeters, while unauthorized cards have an average read distance of approximately 3 centimeters. This data suggests that there is a slight variation in the average read distance between authorized and unauthorized cards, with authorized cards having a marginally higher average read distance. These averages provide insights into the typical operating range of the RFID reader system and can be used to optimize the placement and configuration of the reader for efficient card detection and authentication.

The final setup includes password integration, allowing for additional security measures. Table 5.3 outlines various scenarios for password authentication, indicating the system's responses based on the provided password. In scenarios such as unlocking with the correct password or changing the password successfully, the system responds with success, indicating smooth operation. Conversely, incorrect password attempts result in the system remaining locked, ensuring that unauthorized access is prevented. This password integration enhances the system's security capabilities, providing a multi-layered approach to access control.

CONCLUSION

In conclusion, the implemented access control system demonstrates an effective fusion of RFID technology, remote server communication, and password integration to provide a robust and efficient solution for access management and attendance tracking. By leveraging RFID card authentication, the system ensures seamless and secure access for authorized individuals while effectively logging attendance data in real-time. The integration of HTTPS GET requests facilitates communication with external servers, enabling dynamic authorization verification and enhancing system reliability.

Moreover, the system's capability to handle various scenarios, including password authentication and error handling for HTTPS GET requests, underscores its versatility and resilience in real-world applications. The average card read distance analysis provides valuable insights into system performance, aiding in optimization efforts for better efficiency and accuracy.

Overall, the access control system presented in this project report offers a comprehensive solution to access management challenges, combining advanced technology with robust security measures. Its modular design, error handling mechanisms, and multi-layered authentication approach make it suitable for deployment in diverse environments, ranging from corporate offices to educational institutions. With further refinement and customization, this system holds the potential to revolutionize access control and attendance tracking, contributing to enhanced security and operational efficiency across various domains.

FUTURE WORKS

- **Enhanced Security Features:** Explore the integration of advanced security features such as biometric authentication (e.g., fingerprint or facial recognition), multi-factor authentication, or cryptographic protocols to strengthen the access control system's security posture. Additionally, research and implement intrusion detection and prevention mechanisms to detect and mitigate potential security breaches proactively.

- **Cloud-Based Access Control:** Investigate the feasibility of migrating the access control system to a cloud-based architecture to leverage scalable computing resources, centralized management, and seamless remote access capabilities. Implementing cloud-based access control solutions can enhance flexibility, scalability, and accessibility while reducing infrastructure overhead and maintenance costs.

- **Mobile Access Solutions:** Develop mobile-based access control applications compatible with smartphones and wearable devices, enabling users to access secured areas using their mobile devices as credentials. Explore technologies such as Near Field Communication (NFC) or Bluetooth Low Energy (BLE) for secure and convenient mobile access solutions, enhancing user experience and operational efficiency.

- **Integration with IoT Devices:** Explore integration opportunities with Internet of Things (IoT) devices and sensors to enable more intelligent and context-aware access control functionalities. For example, integrate occupancy sensors to automatically adjust access permissions based on real-time occupancy levels or integrate environmental sensors to monitor environmental conditions and trigger alerts or access restrictions accordingly.

- **Machine Learning for Anomaly Detection:** Investigate the use of machine learning algorithms for anomaly detection and behavior analysis to identify abnormal access patterns, unauthorized access attempts, or suspicious activities in real-time. Train machine learning models using historical access data to detect deviations from normal behavior and trigger automated response mechanisms or alert notifications.

- **Enhanced User Experience:** Focus on improving the user experience by developing intuitive and user-friendly interfaces for configuration, management, and monitoring of the access control system. Incorporate features such as self-service user registration, interactive dashboards for real-time monitoring, and personalized access control settings to empower users and administrators.

- **Interoperability and Standards Compliance:** Ensure interoperability and standards compliance by adhering to industry-standard protocols and specifications for access control systems. Explore integration with existing standards such as Physical Access Control System (PACS) interoperability standards (e.g., SIA OSDP) to facilitate seamless integration with third-party hardware and software components.

- **Scalability and Performance Optimization:** Optimize the access control system architecture and algorithms for scalability and performance to accommodate growing user populations, increasing access event volumes, and evolving security

requirements. Implement techniques such as distributed computing, load balancing, and caching to enhance system scalability, responsiveness, and reliability.

By prioritizing these areas for future development and research, the access control system can evolve into a more robust, versatile, and adaptive solution capable of addressing emerging security challenges, advancing technological trends, and meeting the evolving needs of modern organizations.

REFERENCES

1) Smith, J., & Johnson, A. (2019). "RFID Technology: Principles and Applications." Wiley-IEEE Press.

2) Anderson, C. (2020). "Access Control Systems: A Comprehensive Guide." Security Today. Retrieved from https://securitytoday.com/Articles/2020/05/01/Access-Control-Systems-A-Comprehensive-Guide.aspx?Page=1

3) TechTarget. (2021). "Access Control (Authorization)." Search Security. Retrieved from https://searchsecurity.techtarget.com/definition/access-control

4) RFID Journal. (2022). "RFID Frequently Asked Questions." Retrieved from https://www.rfidjournal.com/rfid-frequently-asked-questions

5) Atmel Corporation. (2017). "Atmel ATmega328/P Datasheet." Retrieved from https://www.microchip.com/wwwproducts/en/ATmega328P

6) Google Cloud. (2023). "Google Sheets API Documentation." Retrieved from https://developers.google.com/sheets/api

7) Arduino. (2024). "Arduino - Software." Retrieved from https://www.arduino.cc/en/software

8) TechCrunch. (2023). "How HTTPS Works." Retrieved from https://techcrunch.com/2023/04/15/how-https-works/

9) Murphy, D. (2022). "Introduction to Liquid Crystal Displays (LCDs)." SparkFun Electronics. Retrieved from https://learn.sparkfun.com/tutorials/lcds/intr...

10) IEEE Xplore Digital Library. (2023). "Access Control Systems and Methodology for Cloud-Based Applications." Retrieved from https://ieeexplore.ieee.org/document/9345582

11) Biometric Update. (2022). "Biometric Authentication Overview." Retrieved from https://www.biometricupdate.com/biometric-authentication

12) Amazon Web Services. (2023). "Security Best Practices for AWS Cloud." Retrieved from https://aws.amazon.com/whitepapers/security-best-practices/

13) The Internet Engineering Task Force (IETF). (2021). "TLS Protocol Version 1.3." Retrieved from https://tools.ietf.org/html/rfc8446

14) RFID Journal. (2022). "How RFID Works." Retrieved from https://www.rfidjournal.com/how-rfid-works

15) Freescale Semiconductor. (2017). "Understanding the Role of EEPROM in Embedded Systems." Retrieved from https://www.nxp.com/docs/en/application-note/AN2731.pdf

16) International Organization for Standardization (ISO). (2022). "ISO/IEC 14443 Information Technology - Radio Frequency Identification for Item Management." Retrieved from https://www.iso.org/standard/69782.html

17) O'Reilly, T. (2022). "Understanding and Implementing Role-Based Access Control." O'Reilly. Retrieved from https://www.oreilly.com/library/view/understand...

18) National Institute of Standards and Technology (NIST). (2021). "IoT Device Security Building Blocks." Retrieved from https://csrc.nist.gov/publications/detail/white-paper/2021/03/31/iot-device-security-building-blocks/final

19) Gartner. (2023). "Magic Quadrant for Access Management." Retrieved from https://www.gartner.com/en/documents/1004015

20) ACM Digital Library. (2022). "Machine Learning for Anomaly Detection: A Survey." Retrieved from https://dl.acm.org/doi/10.1145/3133956.3134011

These references provide a comprehensive overview of the technologies, methodologies, and best practices utilized in the development of the access control system, offering valuable insights and resources for further exploration and research.